DDPM derivations

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March 2025

1 Problem setup

Let x_0 be the original image, x_t be the image of step t, and x_T be the final Gaussian noise.

We have two processes, namely, forward and reverse. We denote the forward process as $q(x_t|x_{t-1})$ and the reverse process as $p(x_{t-1}|x_t)$. Here we are trying to learn the reverse process.

2 Forward Process

Note forward process is a Markov chain that we gradually add Gaussian noise to the data. Thus, we can write the forward process as

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{1}$$

where β_t is the variance scheduler, and $0 < \beta_t < 1$.
We can then use the reparametrization trick:

$$\begin{aligned} q(x_t|x_{t-1}) : &= \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \\ &= \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon \end{aligned}$$

where $\epsilon \sim N(0,1)$.
We let $\alpha_t = 1 - \beta_t$, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, we can further expand $q(x_t|x_{t-1})$ to $q(x_t|x_0)$ as follows:

$$\begin{aligned} q(x_t|x_{t-1}) &= \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon \\ \implies q(x_{t-1}|x_{t-2}) &= \sqrt{\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_{t-1}}\epsilon \\ \implies q(x_t|x_{t-2}) &= \sqrt{\alpha_t}\sqrt{\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t}\sqrt{1-\alpha_{t-1}}\epsilon + \sqrt{1-\alpha_t}\epsilon \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t - \alpha_t\alpha_{t-1}}\epsilon + \sqrt{1-\alpha_t}\epsilon \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t - \alpha_t\alpha_{t-1} + 1 - \alpha_t}\epsilon \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1 - \alpha_t\alpha_{t-1}}\epsilon \end{aligned}$$

Last step because the two ϵ are sampled independently, thus the parameters can be directly added. This way, we can reach all the way to $q(x_t|x_0)$, and it can be written as:

$$q(x_t|x_0) = \sqrt{\alpha_t \dots \alpha_0}x_0 + \sqrt{1 - \alpha_t \dots \alpha_0}\epsilon \tag{2}$$

$$= \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \tag{3}$$

Thus, we can directly model $q(x_t|x_0)$ as $\mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, \sqrt{1 - \bar{\alpha}_t}I)$. In addition, from the original paper, they mentioned that "We ignore the fact that the forward process variances β_t are learnable by reparameterization and instead fix them to constants." One shouldn't be worried about the β_t being learnable parameter here.

3 Reverse Process

Now we model the reverse process as

$$p(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{4}$$

where $\mu_\theta(x_t, t)$ and $\Sigma_\theta(x_t, t)$ are parameters to be learned. In the original paper, the variance $\Sigma_\theta(x_t, t)$ is fixed for simplicity. From paper, they stated that "Experimentally, both $\sigma_t^2 = \beta_t$ and $\sigma_t^2 = \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$ had similar results." We'll not focus on the discussion of the variance in the later part of the analysis, and one can check the subsequent study conducted by OpenAI for more details on the learning of the variance.

Similar to most of the similar tasks, we start with negative log likelihood $-\log(p_\theta(x_0))$. One can note that this is hard to directly access, as it will depends on all the t steps before. Thus, we can model this through usual variational bound on negative log likelihood (just the reverse of the ELBO, but the same concept):

$$\begin{aligned} -\log(p_\theta(x_0)) &\leq -\log(p_\theta(x_0)) + D_{KL}(q(x_{1:T}|x_0)||p_\theta(x_{1:T}|x_0)) \\ &= -\log(p_\theta(x_0)) + \log(\frac{q(x_{1:T}|x_0)}{p_\theta(x_{1:T}|x_0)}) \end{aligned}$$

Now we apply the Bayes' rule on the denominator:

$$p_\theta(x_{1:T}|x_0) = \frac{p_\theta(x_0|x_{1:T})p_\theta(x_{1:T})}{p_\theta(x_0)} = \frac{p_\theta(x_{0:T})}{p_\theta(x_0)}$$

Now we substitute this term back to the equation and get:

$$\begin{aligned} -\log(p_\theta(x_0)) + \log(\frac{q(x_{1:T}|x_0)}{p_\theta(x_{1:T}|x_0)}) &= -\log(p_\theta(x_0)) + \log(\frac{q(x_{1:T}|x_0)}{\frac{p_\theta(x_{0:T})}{p_\theta(x_0)}}) \\ &= -\log(p_\theta(x_0)) + \log(\frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})}) + \log(p_\theta(x_0)) \\ &= \log(\frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})}) \\ &\implies -\log(p_\theta(x_0)) \leq \log(\frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})}) := L \end{aligned}$$

We note that the numerator is just the forward process, and now we expand the denominator as $p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$. We then substitute this back to L and get:

$$\begin{aligned}
L &= \log(\frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})}) \\
&= \log(\frac{q(x_{1:T}|x_0)}{p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)}) \\
&= -\log(p(x_T)) + \log(\frac{\prod_{t=1}^{T} q(x_t|x_{t-1})}{\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)}) \\
&= -\log(p(x_T)) + \sum_{t=1}^{T} \log(\frac{q(x_t|x_{t-1})}{p_\theta(x_{t-1}|x_t)}) \\
&= -\log(p(x_T)) + \sum_{t=2}^{T} \log(\frac{q(x_t|x_{t-1})}{p_\theta(x_{t-1}|x_t)}) + \log(\frac{q(x_1|x_0)}{p_\theta(x_0|x_1)})
\end{aligned}$$

Note here we extract the first step out for later steps.

we apply the Bayes' Rule to the forward step in the numerator and get:

$$q(x_t|x_{t-1}) = \frac{q(x_{t-1}|x_t)q(x_t)}{q(x_{t-1})}$$

For this equation, one should notice that the variance is high when t is large. One can imagine that for a highly noisy image, it's hard to know which step we are at. Thus, we additionally condition these terms on x_0, AKA original image, to reduce the variance. We then substitute this term back in L and get:

$$-\log(p(x_T)) + \sum_{t=2}^{T} \log(\frac{q(x_t|x_0)q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)q(x_{t-1}|x_0)}) + \log(\frac{q(x_1|x_0)}{p_\theta(x_0|x_1)})$$

Note that if we don't take the first step out, we'll end up getting $q(x_1|x_0) = \frac{q(x_0|x_1,x_0)q(x_1|x_0)}{q(x_0|x_0)}$, which doesn't make much sense.
We then split the equation and get:

$$-\log(p(x_T)) + \sum_{t=2}^{T} \log(\frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)}) + \sum_{t=2}^{T} \log(\frac{q(x_t|x_0)}{q(x_{t-1}|x_0)}) + \log(\frac{q(x_1|x_0)}{p_\theta(x_0|x_1)})$$

For the term $\sum_{t=2}^{T} \log(\frac{q(x_t|x_0)}{q(x_{t-1}|x_0)})$, we can expand this and get:

$$\sum_{t=2}^{T} \log(\frac{q(x_t|x_0)}{q(x_{t-1}|x_0)}) = \log(q(x_t|x_0)) - \log(q(x_{t-1}|x_0)) + \log(q(x_{t-1}|x_0)) - \log(q(x_{t-2}|x_0)) + \ldots - \log(q(x_1|x_0))$$

One should note that all the intermediate terms will cancel out and we are left with the first and the last terms, and thus this whole summation will collapse as:

$$\sum_{t=2}^{T} \log(\frac{q(x_t|x_0)}{q(x_{t-1}|x_0)}) = \log(\frac{q(x_T|x_0)}{q(x_1|x_0)})$$

We again substitute these terms back to L, and we get:

$$\begin{aligned}
L &= -\log(p(x_T)) + \sum_{t=2}^{T} \log(\frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)}) + \log(\frac{q(x_T|x_0)}{q(x_1|x_0)}) + \log(\frac{q(x_1|x_0)}{p_\theta(x_0|x_1)}) \\
&= \log(\frac{q(x_T|x_0)}{p(x_T)}) + \sum_{t=2}^{T} \log(\frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)}) - \log(p_\theta(x_0|x_1)) \\
&= D_{KL}(q(x_T|x_0)||p(x_T)) + \sum_{t=2}^{T} D_{KL}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t)) - \log(p_\theta(x_0|x_1))
\end{aligned}$$

Examining the three terms left, we can observe that for the first term, there's no parameter to learn, $p(x_T)$ is just random Gaussian noise, and $q(x_T|x_0)$ is the forward process, which should be similar enough to Gaussian noise as well by setting. Therefore, the distance between the two distributions shouldn't differ by much as they are both Gaussian noise, and we may ignore this term.

For the second term, we have the same form for both forward and backward processes. It can be shown that the numerator can be modeled as $\mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$, and $p_\theta(x_{t-1|x_t})$ can be modeled as $\mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \beta I)$, where β is fixed as discussed.

For the numerator, $\tilde{\beta}_t$ was derived before, and $\tilde{\mu}_t$ can be shown that:

$$\tilde{\mu}_t = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \frac{\beta_t}{1 - \sqrt{\bar{\alpha}_t}\epsilon})$$

This is equivalent to subtracting scaled random noise from x_t. For more details, please find the original DDPM paper.

To learn μ_θ, we use an MSE loss defined as:

$$\frac{1}{2\sigma_t^2} \| \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \frac{\beta_t}{1 - \sqrt{\bar{\alpha}_t}}\epsilon) - \mu_\theta(x_t, t) \|^2 \tag{5}$$

Since x_t is already known, we can instead learn $\epsilon_\theta(x_t, t)$:

$$\frac{1}{2\sigma_t^2} \| \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \frac{\beta_t}{1 - \sqrt{\bar{\alpha}_t}}\epsilon) - \mu_\theta(x_t, t) \|^2 \tag{6}$$

$$\implies \frac{1}{2\sigma_t^2} \| \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \frac{\beta_t}{1 - \sqrt{\bar{\alpha}_t}}\epsilon) - \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \frac{\beta_t}{1 - \sqrt{\bar{\alpha}_t}}\epsilon_\theta(x_t, t)) \|^2 \tag{7}$$

$$\implies \frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \tag{8}$$

Empirically, without the scaling term even gives better sampling quality and easier implementation. Thus, this leads to our final MSE loss:

$$\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \tag{9}$$

This formula means that it's sufficient that we only predict the noise for each step to learn the whole reverse process.